HUMAN CENTRIC COMPUTING REPORT

QualiaDB / Webizen
Agency, Wellbeing and Cooperative Capability

Project intent, functional design and technical capability across the hierarchy of human needs

A concept and design report for the QualiaDB / Webizen ecosystem

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Peace infrastructure for the natural person - running on your hardware, connected to the world on your terms.

Reader's map

This report explains the project as an intended Human Centric computing platform. It starts with the human purpose, uses Maslow's hierarchy as a practical design lens, then traces how rights, supported agency, mindware, cooperation and fair exchange are translated into technical architecture.

- Part I - Human purpose. A needs-based account of wellbeing, safety, belonging, esteem, self-development and collective contribution.
- Part II - Rights and agency. Personhood, selfhood, consent, guardianship, accessibility, accountability and control of personal mindware.
- Part III - Technical capability. Semantic data, logic, AI, scientific computing, geometry, privacy, identity, networking and application surfaces.

- Part IV - Human journeys and outcomes. Concrete scenarios and an outcome framework for evaluating whether the platform advances people's interests.

SCOPE This is an intent and design report, not a release-status audit, medical device claim, legal opinion or assertion that technology can satisfy human needs by itself. The report uses 'designed to', 'seeks to' and 'positions' deliberately: social legitimacy depends on accountable deployment, competent care, law, culture and the person's own choices.

Executive overview

QualiaDB is conceived as a local-first semantic and computational foundation for Webizen: a personal computing environment in which data, AI, formal reasoning and networked cooperation are arranged around the interests of the natural person who owns and uses the system. Its central design proposition is fiduciary rather than extractive: software should act as an instrument of the person, should make its authority legible, and should not quietly convert identity, health, relationships or creative work into platform inventory. [P1]

The platform joins capabilities that are usually fragmented across cloud services. A compact semantic graph provides durable memory and context. Formal logic expresses obligations, permissions, knowledge, uncertainty, conflicts, time, delegation and responsibility. Local AI and machine learning assist interpretation and creation. Scientific, geometric and data-science libraries support modelling. Identity, credentials, provenance and consent constrain who may act, on what basis and for what purpose. QApps and cooperative records turn these capabilities into practical workflows for health, welfare, learning, research, projects and economic participation.

Maslow's theory of motivation is used here as an organizing heuristic, not as a rigid clinical sequence. Human needs overlap, vary by culture and circumstance, and may be pursued simultaneously. The useful design question is therefore not 'Which layer has been completed?' but 'What capabilities, safeguards and relationships help a person pursue needs at each layer without sacrificing rights at another?' [E1]

CORE INTENT To make advanced computation serve human agency: private enough for inward life, rigorous enough for consequential decisions, interoperable enough for institutions, cooperative enough for shared work, and economically expressive enough to recognize contribution without converting the person into a commodity.

- Physiological and welfare support: a personal vault for health, biometrics, medications, sleep, clinical records, finance, housing and assistance pathways.
- Safety and stability: local control, sanctuary-class information boundaries, cryptographic identity, provenance, scoped access, deterministic execution and supported decision safeguards.
- Belonging and social wellbeing: relationship-aware records, care networks, communication, guardianship commons, social cooperatives and peer-to-peer exchange.
- Esteem and agency: authorship, credentials, inspectable reasoning, contribution records, useful ICT tools and control over how one's work and representations are used.

- Self-development and collective contribution: person-controlled mindware, research and modelling workbenches, local AI, scientific computation, QApps, shared knowledge and permissive commons.

Part I - Human purpose through a needs-based lens

The project's purpose can be read as a layered capability system. Lower-order needs require reliable records, privacy, care coordination and material security. Relational needs require trustworthy communication and shared context. Esteem and self-development require authorship, competence, creative tools and control. Across every layer, the governing unit is the person-in-relation: an individual with an inward life, social roles, duties, dependencies, communities and evolving projects.

1. Physiological wellbeing, health and welfare

At the most immediate layer, Webizen is designed to help a person assemble a longitudinal, person-controlled account of the conditions that shape daily life. The WellFair domain models weight, sleep, heart rate, steps, medications, diet, clinical documents, conditions, allergies, mental wellbeing observations, welfare cases, housing-safety reports, finance and assistance streams. These records are not treated as isolated app silos: semantic translation, temporal context, evidence status and sensitivity classification make them available for cross-domain reasoning under the person's control. [P2]

The intended value is continuity. A person can compare observations across time, prepare for consultations, identify missing evidence, understand the relationship between health and material circumstances, and decide what to disclose. The anatomy and body-system model adds a structured way to connect observations, burdens, pathways and scorecards without reducing the person to a single score. Clinical formulas and rule systems are supporting instruments; consequential interpretation remains attributable and reviewable.

- Personal health memory: structured observations, documents and provenance that remain available even when providers or platforms change.
- Welfare coordination: assistance needs, benefits streams, case tasks, government letters and practical support are represented alongside health and life context.
- Privacy-preserving analysis: local computation, sensitivity classes, differential privacy and homomorphic methods are intended to enable useful analysis with less disclosure.
- Accessible self-understanding: visualization, plain-language interfaces and multimodal records aim to help the person participate in decisions about their own body and circumstances.

2. Safety, continuity and freedom from coercive dependence

Safety is addressed as more than cybersecurity. It includes physical and relational safety, predictable access to one's records, protection from data extraction, continuity during institutional change, and the ability to seek support without surrendering control. Local-first storage and execution reduce reliance on a remote platform. Sanctuary and bilateral micro-commons classifications are intended to prevent sensitive medical, legal, guardianship and inward-self data from becoming public-addressable assets. [P1][P3]

The system's bounded execution model is also a safety design. Fixed-size semantic records, caller-owned buffers, deterministic evaluators and explicit memory ceilings make resource use more predictable across personal computers, browsers and edge devices. Cryptographic identities, capability profiles, verifiable credentials, governed writes and provenance records are intended to answer: who acted, under what authority, on which data, for what purpose, and with what result?

Safety also requires exit and reversibility. The project's checkpoint, branch, revocation and consent concepts support the idea that people should be able to review, correct, withdraw, fork, archive or selectively disclose their digital history. A secure system that cannot be understood or exited can still become coercive.

3. Belonging, care and social wellbeing

Human wellbeing is relational. The project therefore treats family, carers, peers, professionals, communities and collaborators as structured relationships rather than a contact list. SocialWebNet, Solid/LDP interoperability, live-share agreements, guardianship records and cooperative projects are designed to support communication and coordinated action without placing a central platform in the middle of every relationship.

A key sociological feature is the distinction between support and control. Guardianship is modelled as scoped stewardship or supported agency, potentially involving multiple participants, role separation, consent, triggers, provenance and M-of-N authorization. The purpose is to amplify the principal's ability to act, not to treat disability, age, crisis or dependence as a basis for erasing personhood. This aligns with the human-rights direction from substitute decision-making toward support that respects the person's will and preferences. [E4][P4]

- Shared context without forced centralization: people and institutions can exchange semantic records through open protocols and peer relationships.
- Care as accountable relationship: supporters act within named domains, purposes, time windows and revocable scopes.
- Community memory: projects, cultural resources and shared ontologies can preserve contribution, provenance and differing perspectives.
- Conflict without social explosion: paraconsistent and defeasible logic provide ways to preserve disagreement and uncertainty rather than flattening it into a single authoritative narrative.

4. Esteem, competence and effective agency

Esteem in a computing platform is not a badge or engagement score. It is the practical ability to understand one's situation, produce work, be recognized as its author, develop competence, negotiate terms and have contributions counted. QualiaDB's provenance, credentials, contribution records and project ledgers are designed to preserve the difference between the natural person, the role they occupy, the software instruments they use and the legal entities with which they interact.

Useful ICT matters here. The intended experience is not limited to consuming entertainment or receiving algorithmic recommendations. A Webizen user should be able to query knowledge, build ontologies, analyse data, create documents and models, write or run QApps, conduct research, coordinate tasks, manage a budget, evaluate evidence and publish attributable work. Local AI assists these activities while remaining one instrument among others.

5. Self-development, creativity and self-actualization

At the growth-oriented layer, the platform becomes a workbench for inquiry and creation. The person can build a personal knowledge graph, explore hypotheses, model systems, learn from local or shared corpora, combine symbolic and statistical methods, and develop projects with others. The broad scientific libraries are relevant because self-development is domain-specific: a person may be exploring biology, engineering, finance, language, geometry, law, art, civic life or a new field that spans several of them.

The platform's 'mindware' concept is central. It describes the structured memory, models, agents, preferences, goals, methods and semantic context that extend a person's capacity to think and act. Mindware is designed to remain inspectable, portable and governed by the person. It is not a claim that software is the person's mind, that a model has equal personhood, or that inferred psychological attributes should be available for covert scoring.

6. Beyond the five levels: contribution and collective flourishing

Maslow's familiar five-level presentation can be extended with a collective question: how can personal growth contribute to shared knowledge without requiring uncompensated extraction? The Permissive Commons and Threshold Shift concepts seek to combine distribution, cultural and linguistic plurality, creator recognition, bounded economic obligations and eventual shared availability. The intended outcome is a commons that can reward the labour and risk involved in creating useful resources while resisting perpetual rent-seeking. [P3]

Table 1. Human needs translated into platform objectives

Needs lens	Human objective	Designed platform support	Rights test
Physiological	Continuity of health, care and material support	WellFair records, anatomy, medication, sleep, clinical and welfare tooling	Does the person control access, context and correction?
Safety	Privacy, stability, protection and predictable authority	Local-first Q42 storage, sanctuary classes, credentials, capability gates, provenance	Can use be limited, audited, revoked and remedied?
Belonging	Trustworthy relations and cooperative care	Peer exchange, live-share agreements, guardianship commons, social cooperatives	Does support preserve will, preferences and equal personhood?
Esteem	Competence, recognition, authorship and fair terms	Research tools, credentials, contribution ledgers, project and economic logic	Are labour, authorship and bargaining position legible?
Self-development	Learning, creativity, inquiry and purposeful work	Mindware workbench, AI/ML, logic, scientific modelling, QApps	Can the person inspect, adapt, export and govern the instruments?
Collective	Shared knowledge and institutions that do not enclose people	Permissive Commons, provenance, interoperable standards, threshold licensing	Are communities, cultures and contributors protected from extraction?

Part II - Human rights, personhood and control

7. Rights are architectural requirements

The project treats human rights as design inputs to data structures, authority models and execution gates. The relevant rights span privacy, health, social security, work, fair remuneration, participation in cultural and scientific life, freedom of association, equal recognition before the law, accessibility and effective remedy. The Universal Declaration of Human Rights and the International Covenant on Economic, Social and Cultural Rights connect dignity with privacy, social protection, health, work and just conditions; the Convention on the Rights of Persons with Disabilities adds a particularly important standard for legal capacity and supported decision-making. [E2][E3][E4]

This means the system should not merely display a policy notice. It should be able to represent purpose, consent, authority, time, sensitivity, delegation, revocation, provenance, conflict and duties in computable form. Formal logic does not replace law or human judgment; it makes assumptions and constraints more explicit, testable and contestable.

8. Natural person, personhood, role and instrument

QualiaDB's ontological design distinguishes four concepts that many platforms blur: the natural person who bears dignity and rights; personhood as the person's socio-legal ability to act and be recognized; roles such as guardian, clinician, contributor or officer; and instruments such as software agents, models and legal entities. The distinction is intended to prevent category errors in which an artificial agent is treated as a rights-bearing peer or a corporation's interests silently override the human principal. [P5]

Selfhood is used for the most inward and inalienable aspects of the person: body, biometrics, genetic and reproductive attributes, identity, conscience, intimate memory and the mindware that can act as or unlock the person. Personhood covers outward social and legal participation. A person may delegate tasks or authority within personhood, but selfhood is not delegated by default. Disclosure is a deliberate manifestation from the inward sphere to the relational sphere, not a permanent transfer of ownership.

9. Supported agency and guardianship

The guardianship and supported-agency model is designed around a simple test: does the arrangement increase the principal's practical ability to pursue their own rights and preferences? Authority is decomposed by domain, trigger, role, evidence, time and accountability. Medical support need not imply financial authority; crisis activation need not become permanent control; a carer need not become the sole decision-maker; and a software recommendation never becomes authority merely because it is computationally sophisticated.

- Presumption of personhood and capacity: support begins from equal recognition, not from a deficit label.
- Will and preferences: decisions should follow the principal's known wishes, values and communication methods, with uncertainty made visible.

- Scope and proportionality: delegated authority is limited to the minimum domain, purpose, action and time required.
- Plural support and separation of powers: high-risk actions may require multiple participants, independent review or M-of-N approval.
- Revocation and restoration: support relationships and crisis powers must be reviewable, time-bounded and capable of ending.
- Provenance and remedy: the system records who relied on what evidence and creates a path for dispute, correction and accountability.

GUARDIANSHIP PRINCIPLE Support is successful when it expands the person's agency. A technically valid authorization that suppresses the person's will, hides conflicts of interest or becomes impossible to revoke is a failure of the Human Centric design.

10. A rights-centred design covenant

Commitment	What it means in system design	Illustrative mechanisms
Agency	The person sets purposes and can understand or contest consequential actions.	Intent frames, local controls, explainable provenance, human review
Privacy & sanctuary	Inward and sensitive data are not public assets or default network payloads.	Local-first storage, sensitivity classes, encryption, disclosure gates
Consent	Permission is specific, informed, attributable, time-aware and revocable.	Consent credentials, usage agreements, capability scopes, revocation
Non-discrimination	Models and rules must not turn health, disability, culture or social position into hidden exclusion.	Provenance, uncertainty, protected classifications, review and audit
Accessibility	The person can perceive, understand and operate the system using appropriate modalities.	Multimodal semantics, adaptable presentation, language support, browser/edge profiles
Accountability	Authority and contribution do not disappear into 'the algorithm'.	DIDs, signatures, WAL/provenance, actor/role separation, conduct records
Fair participation	Useful work can be recognized, negotiated and compensated without endless enclosure.	Contribution ledgers, obligations, escrow, cooperative governance, threshold shift
Remedy & exit	People can correct, revoke, export, branch, archive or leave.	Versioning, checkpoints, dispute states, portable RDF/CBOR-LD, interoperable identity

11. Mindware as a person-controlled cognitive extension

In this project, mindware can be understood as the computational apparatus a person uses to remember, interpret, decide, create and coordinate. It may include a personal knowledge graph, authored concepts, observations, goals, commitments, preferences, evidence, timelines, models, prompts, tools, agents, learned representations and the procedures by which these elements are combined. It is socio-neuromorphic in the broad sense: designed around human cognition-in-relation, not around an isolated model pretending to be the person.

Advanced modelling of functional attributes is therefore possible without treating a person as a fixed psychological profile. The system can represent attention demands, memory supports, communication preferences, confidence, source quality, competing beliefs, decision dependencies, stressors, routines, skills, goals and support relationships as contextual claims. Epistemic status, provenance and time allow the platform to distinguish what the person stated, what an instrument observed, what a model inferred and what remains disputed.

The control model is as important as the model itself. Person-controlled mindware should be inspectable, selectively shareable, correctable, versioned, portable and revocable. Sensitive inferences should be purpose-bound. A model that acts for the person should carry limited capabilities and attributable provenance. A guardian or institution should receive only the slice required for the agreed purpose. The person should be able to preserve private inward material even while using outward personhood tools.

- Memory: durable, semantic and temporal records rather than a provider-owned chat history.
- Reasoning: explicit rules, uncertainty and competing interpretations alongside statistical prediction.
- Learning: reusable tools, datasets, ontologies and models that can be adapted to the person's aims.
- Expression: documents, visualizations, QApps, code, research and cooperative artefacts with retained authorship.
- Action: scoped agents and workflows that can query, calculate, communicate and transact within granted authority.
- Reflection: scorecards and self-assessments held as protected selfhood content, not converted into external reputation scores.

12. Cooperative projects and fair economic support

QualiaDB is designed to help people move from private capability to shared undertaking. A project can have a durable identity, goals, members, roles, tasks, evidence, contributions, costs, decisions and changing rules. Event-sourced work items and provenance make collaboration replayable and auditable. Add-wins contribution records reduce the risk that concurrent edits erase labour. Shared ontologies make terms and expectations explicit across participants.

Economic support is treated as part of agency rather than an afterthought. The WellFair ledger can represent personal finance and derived balances. Cooperative project records can track effort and capital. Deontic and jural logic can express obligations, claims, permissions, liabilities and unmet duty-bearers. Value-flow, escrow and threshold licensing concepts provide a vocabulary for payment, risk, contribution and the transition of a resource into a commons after a bounded obligation has been met.

Fairness is not produced by arithmetic alone. The design must preserve bargaining context, skill, responsibility, risk, care work, cultural stewardship and the effect of work on the person's life. The UN Committee on Economic, Social and Cultural Rights notes that fair remuneration is broader than a minimum wage and should consider responsibilities, skill, hardship, health and family life. The project's semantic approach is well suited to representing those factors, provided they remain contestable and are not used for discriminatory scoring. [E5]

An intended cooperative lifecycle

1. Form the project: define the project's DID, purpose, values, constituencies, governance and capability boundaries.
2. Agree the work: express tasks, roles, evidence, deadlines, dependencies, permissions and obligations in human- and machine-readable form.
3. Perform and record: use QApps, data tools, models and agents while preserving actor, source, transformation and contribution provenance.
4. Review and resolve: use consensus, defeasible logic, dispute states and human review to address conflict without erasing minority evidence.
5. Compensate and distribute: derive obligations, settle agreed value flows, respect licences and consent, then publish or share under appropriate commons terms.
6. Evolve without amnesia: checkpoint, branch, merge and preserve the derivative chain so later work can recognize earlier contributors and decisions.

Part III - Technical capabilities that serve the human purpose

The platform's breadth is intentional. Human goals do not respect software product boundaries: a housing problem may be legal, financial, geographic, health-related and relational at once. QualiaDB therefore combines semantic interoperability with multiple forms of computation. The following sections describe the intended capability system as an integrated personal-computing substrate.

13. Semantic graph, ontology and durable context

The core semantic datum is the 48-byte NQuin, a compact six-field record for subject, predicate, object, context, metadata and parity. Q42 volumes organize these records for deterministic, memory-mapped and compressed storage. Context fields, sensitivity classes, clocks and routing metadata make the graph suitable for more than facts: it can represent worlds, contracts, provenance, access lanes and temporal states. [P1][P6]

RDF and linked-data compatibility provide an open model for identity and knowledge. SPARQL supports querying across people, events, resources, graphs and time; RDF-Star-style quoted statements allow claims about claims; SHACL and ShEx-oriented validation constrain structure; N3 rules express implications; and OWL/description-logic materialization supports class relationships. Standard serializations and a Solid/LDP bridge reduce lock-in and make it possible to exchange data with institutions that do not run the full stack. [E6][E7][E8][E9]

- Durable identifiers: DIDs and hashed IRIs separate identity from a single provider's domain or database key.
- Contextual truth: named graphs, provenance and epistemic status distinguish assertion, evidence, interpretation and world-view.
- Temporal continuity: clocks, intervals, snapshots and trace logic support changing circumstances and retrospective questions.

- Portability: RDF, JSON-LD, CBOR-LD, Turtle, N-Quads and standard query results support movement across tools and institutions.

14. Native formats as enabling infrastructure

Open standards remain the public language of the platform, but a system designed for continuous reasoning on personal devices must also solve a different engineering problem: bounded memory, predictable execution, offline custody, provenance, high-density geometry and locally resident model weights. The project therefore defines a family of purpose-specific native formats. They are implementation substrates beneath an interoperability surface, not proprietary replacements for RDF, Solid or the web. [P7][P8][P9]

Why interchange formats are not sufficient on their own

RDF, Turtle, JSON-LD, HTTP and Solid are valuable because they make meaning and network behaviour intelligible across organizations. They are optimized for description, exchange and extensibility. The hot path of a personal computing engine has additional constraints: it may need to evaluate millions of semantic records, traverse geometry, verify provenance or bind model tensors without allocating a large object graph or repeatedly parsing strings. Expanding every vertex, tensor cell or model weight into triples would make the information theoretically open but operationally unusable on affordable hardware.

Native formats turn critical invariants into physical properties that can be inspected and tested: fixed record widths, declared byte order, bounded block sizes, explicit magic and versions, page-aligned sections, relative offsets, checksums, content hashes and fail-closed validation. That makes deterministic CPU, GPU, WASM and edge execution possible while keeping the semantic identity, policy and provenance of an asset in a linked-data control plane.

FORMAT PRINCIPLE A native format is justified only when it adds a capability that an interchange representation cannot provide efficiently or safely, and only when the person retains a documented path to inspect, migrate and export the information.

Native format	Primary responsibility	Human and technical rationale
Unified v3 .q42	Semantic graph and durable context	A self-contained, memory-mappable volume of 48-byte NQuins with embedded lexicon and indexes, block-local compression, temporal/history sections, publication classification and cryptographic roots.
.qualia	Portable vault entry manifest	A Human Centric collection descriptor that names the person's vault context, identifiers, included Q42 volumes, capability envelopes and preferred shell without turning the vault into application-owned data.
.qchk	Capability envelope	A distinct, machine-recognizable place for authority and permitted operations, preventing data containers, application packages and generic checkpoint files from being mistaken for grants of power.
.p64	Local model-weight container	A cache-line and page-aligned, memory-mappable artifact for model tensors, tokenizer data, hyperparameters, manifold coordinates and integrity checks, separated from the semantic truth and provenance graph.
.10d / Tensor10D	Dense geometry and multidimensional state	Compiled mesh, topology, spatial-index and tensor sections that remain compact and deterministic while Q42 records describe identity, units, sensitivity, provenance, fidelity and assurance.

Native format	Primary responsibility	Human and technical rationale
yaml-ld-q42 + qapp.json	Human-authorable workspace and package declaration	Separate presentation/layout semantics from launch metadata, capability requirements and host integration so interfaces can be portable, declarative and least-privileged.
.hcf / .hmc	Semantic hypermedia and verifiable bundles	Combine RDF-native documents with source, graph, geometry, model and media assets in content-addressed packages that can be streamed, verified and projected into conventional web or document formats.

A semantic control plane and a dense execution plane

The formats form a layered architecture. The semantic control plane records identity, meaning, provenance, time, rights, sensitivity, licence and authority. The dense execution plane carries the bytes that must be processed efficiently: compressed graph blocks, model tensors, meshes, topology, spatial indexes and multimodal assets. A content hash, typed reference and validation profile join the two planes. The separation means that a binary asset cannot silently confer authority, while a semantic claim cannot pretend to contain dense data that it merely names.

In the Q42 volume, each NQuin is a fixed 48-byte semantic datum and each 40,960-byte SuperBlock is an independently addressable physical I/O unit. A unified v3 volume adds a 256-byte header, embedded lexicon and indexes, block-local LZ4 compression, optional temporal and Merkle-DAG history sections, and explicit Commons or Sanctuary publication classification. These are not cosmetic optimizations: they enable selective block access, deterministic memory budgets, offline term resolution, verifiable history and a fail-closed boundary between personal material and publishable commons data. [P7]

P64 is intentionally a sibling rather than a subtype of Q42. The semantic graph can say which model was selected, who authorized it, where it came from and which licence or evaluation applies; the P64 file contains the aligned mathematical weights and tokenizer needed for local inference. Likewise, a Q42 geometry manifest carries identity, units, provenance, sensitivity and assurance, while a .10d container carries dense vertices, topology and indexes. This division allows model and spatial computation to be fast without allowing opaque binary payloads to become the source of rights or truth.

Design property	Native mechanism	Human value
Deterministic boundedness	Fixed records, block units, explicit budgets and caller-owned buffers	Reliable operation on personal, mobile and edge devices; reduced dependence on premium cloud infrastructure.
Local custody	Memory-mapped volumes, embedded lexicons and offline-capable manifests	Continuity of access and the ability to use mindware without routine disclosure to a remote provider.
Rights-aware routing	Sanctuary/Commons classification, sensitivity fields and separate capability envelopes	Confidentiality, purpose limitation and authority that travels with the data and derived assets.
Contestable provenance	Parity, CRCs, content hashes, Merkle roots, clocks and history sections	A person can ask what changed, by whom, under which authority and from which source.
Heterogeneous computation	Aligned sections, relative offsets and compact dense assets	Local AI, geometry, simulation and visualization can share a substrate across CPU, GPU, WASM and edge targets.
Exit and reversibility	Versioned formats, migration rules and standards-based projectors	The person can move, inspect or re-express material rather than remaining captive to one interface or runtime.

Interoperability by construction

The intended architecture has two mutually accountable faces. Internally, native files and fixed-layout buffers provide deterministic execution. Externally, RDF datasets and named graphs, SPARQL, SHACL, DIDs, verifiable credentials, PROV-O, Solid/LDP and common media formats provide exchange and institutional compatibility. Translation occurs at an allocation and validation firewall: string-heavy network representations are parsed, bounded and converted into hashes and fixed records before they enter the core; outgoing records are projected back into standards-oriented representations under the person's disclosure policy. [P9][E6][E7][E8][E9][E10]

QISP applies the same principle to immersive and scientific computation. It is designed as a profile over RDF and SPARQL, not a forked query language. RDF names and explains the world; content-addressed assets carry dense meshes and tensors; stable function IRIs invoke bounded native kernels; and provenance and policy describe the result. A client that does not implement QISP can still query ordinary RDF metadata, discover the capability, retrieve authorized alternate representations such as WKT, GML, glTF or descriptor-only RDF, and receive honest errors for unsupported functions. Immersive presentation is never intended to become an accessibility requirement.

Native capability	Open bridge or projection	Interoperability outcome
Q42 graph volumes	RDF 1.2 datasets and named graphs; Turtle, N-Quads, RDF/XML, JSON-LD and CBOR-LD projections; SPARQL query results	Semantics, provenance and context can move without exposing the internal memory layout.
.qualia and yaml-ld-q42	Turtle/N3, JSON-LD or CBOR-LD views; DIDs, credentials and linked-data vocabularies	Vault and workspace descriptions remain inspectable, authorable and portable across shells.
.qchk capability envelopes	Machine-readable DID/credential and policy graphs with explicit scopes	Authority can be interpreted across components without being hidden inside application code or data files.
P64 model containers	GGUF and Safetensors producer profiles, with model identity, licence and evaluation described in Q42	Existing model ecosystems can be imported while execution remains local and provenance remains separate.
.10d and Tensor10D	QISP, GeoSPARQL, QUDT, WKT/GML and glTF/GLB alternate representations	Exact dense computation coexists with ordinary spatial discovery and accessible non-immersive views.
HCF/HMC	HTML, Markdown, PDF, JSON-LD/RDFa and source-asset projectors	A verifiable semantic work can travel through conventional channels without making an emitted view the canonical document.
QApps	PWA/WASM, web components, MCP-style tools and Solid-backed data exchange	Interfaces remain replaceable views over the person's graph and authority rather than becoming the owner of them.

Governance of an evolving format portfolio

Several specifications in the project are deliberately identified as internal, editor or implementer drafts. That maturity should be visible rather than concealed. The architectural objective is a disciplined path from experimentation to dependable public contract: stable magic and version fields, dated vocabulary IRIs, explicit maturity labels, migration notes, conformance classes, published test vectors, independent readers and a deprecation process in which terms are never silently repurposed.

Every native format should therefore carry an interoperability obligation: document its purpose and non-goals; validate lengths, offsets and digests before use; fail closed on unsupported security or policy features; preserve source identity and licence information; provide a bounded migration path; expose accessible and non-immersive representations; and maintain export mappings that can be implemented without the original user interface. This converts format design from an internal performance decision into part of the project's human-rights and anti-lock-in architecture.

INTEROPERABILITY COVENANT Native efficiency must strengthen local agency, not create a new enclosure. A person should be able to understand what an artifact is for, verify which authority applies, migrate it across versions, and export its meaning and usable content through documented open representations.

15. A many-logic reasoning environment

No single logic is adequate for human life. Classical entailment works for some taxonomies, but social and legal situations contain uncertainty, exceptions, conflicting testimony, changing obligations and incomplete evidence. The Webizen VM and modality libraries are designed as a plural reasoning environment in which each logic answers a different kind of question.

Logic family	Question it can express	Human Centric application
Deontic & jural	What is obligatory, permitted or forbidden; who holds the correlative duty, claim, power or immunity?	Agreements, care duties, access terms, creator obligations and rights gaps
Capacity, STIT & responsibility	Could the person act; did an agent see to it; was there a genuine alternative; who is answerable?	Supported decisions, omission, breach, duress, professional and software accountability
Epistemic & doxastic	Who knows or believes what, with which certainty, source and world context?	Clinical evidence, research, testimony, personal knowledge and common understanding
Defeasible & argumentation	Which conclusion holds when rules have exceptions or reasons attack and support one another?	Policy, law, contested evidence, preference and review
Paraconsistent	How can contradictions be isolated without making every conclusion derivable?	Disputed diagnoses, conflicting reports, plural histories and safe intake
Temporal (LTL/CTL/interval)	What must always, eventually or next occur; how do intervals relate?	Care plans, deadlines, monitoring, contracts and provenance timelines
Description & ontology	Is one class subsumed by another; which constraints and relationships follow?	Interoperability, eligibility, classification and semantic forms
Probabilistic, fuzzy & causal	How likely, graded or causally supported is a relationship?	Risk, forecasting, scientific inference and uncertainty-aware support
ASP, dialectical & consensus	Which stable worlds satisfy constraints; how can opposition yield synthesis; what constitutes quorum?	Planning, negotiation, multi-party governance and group decisions

16. Local AI, machine learning and neuro-symbolic assistance

The AI architecture is local-first. Native GGUF and P64 model containers, tokenization, memory-mapped weights, CPU/GPU dispatch and bounded autoregressive decoding are designed to run on the person's hardware. Browser profiles use WebAssembly and WebGPU where appropriate. Model lifecycle, thermal governance and capability profiles allow the same conceptual platform to scale from an edge device to a workstation without assuming unlimited power, memory or network access.

Locality is not the only Human Centric property. The intended inference path combines generative models with symbolic gates, graph context, provenance, grounding and deterministic tools. An LLM can propose language or a plan; the graph supplies attributable context; formal logic checks constraints; and specialized solvers perform calculations that should not be improvised by a language model. This orchestration makes the AI part of a toolchain rather than the sole epistemic authority.

- Language models: local generation, summarization, drafting and conversational access to personal or project knowledge.
- Classical machine learning: model loading, training, optimization, versioning, monitoring and inference interfaces.
- Vision: structured observations, image processing, embeddings, detection interfaces, spatial reconstruction and super-resolution paths.
- Audio: spectral and symbolic processing intended for efficient multimodal interaction without making raw media the default semantic format.
- Neuro-symbolic control: grounding, provenance, intent and output validation connect statistical generation to explicit policy and evidence.
- Tool orchestration: deterministic parsers, solvers and renderers handle tasks where exactness, repeatability or domain-specific computation matters.

17. Scientific, mathematical and data-science capability

QualiaDB includes broad computational libraries so a personal or cooperative project can move from description to analysis. The goal is a common interface in which data, assumptions, methods and results remain connected to semantic context and provenance. This supports advanced technical assistance without forcing the person to surrender their data to a collection of unrelated cloud services.

Capability area	Representative functions	Purpose in a Human Centric platform
Linear algebra & optimization	Dense/sparse operations, decompositions, solvers, privacy-preserving algebra	Foundation for models, statistics, engineering and encrypted computation
Statistics & data science	Datasets, descriptive/inferential analytics, privacy, scheduling, compression	Evidence analysis, trends, uncertainty, responsible release and reproducibility
Symbolic mathematics	Algebra, integration, limits, series, trigonometry, ODEs, polynomial and multivariable calculus	Inspectable derivations, education, engineering and scientific reasoning
Computational geometry	Exact predicates, hulls, Delaunay, meshes, topology, BVH, reconstruction, motion planning, homology	Spatial modelling, anatomy, fabrication, robotics, mapping and scientific visualization
Physics & engineering	Mechanics, fields, thermodynamics, CFD, FEM, vibration, reliability, thermal and structural analysis	Design, simulation, safety analysis and technical learning
Chemistry & biology	Molecular properties, kinetics, quantum chemistry, bioinformatics, protein/k-mer and quantum-biology paths	Research, education, drug/biomolecular analysis and hypothesis support

Capability area	Representative functions	Purpose in a Human Centric platform
Medical computing	Clinical formulas, risk, contraindications, records, imaging DSP and differential-support structures	Clinician/person preparation and computable health context with explicit safeguards
Economics & finance	Ledgers, pricing, risk, portfolio, settlement, value flow, accounting and economic models	Personal planning, cooperative funding, fair exchange and transparent obligations
Quantum interfaces	QUBO/circuit jobs, ground-state resolution and cached solved topologies	Optional acceleration for bounded optimization while retaining classical fallbacks

18. Geometry, Tensor10D and human-readable worlds

Geometry is both a computational library and an organizing metaphor for the platform. Tensor10D coordinates combine quantum context, topology, manifold, spatial position, time and spectral/logical payload. This allows semantic information, simulation state, media and alternative worlds to be projected into a common bounded representation. N-dimensional rendering, depth, bloom, meshes, picking and CPU readback turn complex structures into inspectable scenes.

The human purpose is not novelty for its own sake. Spatial interfaces can help a person explore anatomy, causal relationships, project networks, geographic data, uncertainty and changing states. Exact geometric predicates and deterministic kernels support applications where visual persuasion must not outrun computational correctness.

19. Privacy, security, identity and accountable authority

Privacy is treated as a layered engineering problem: data classification, local custody, cryptography, capability security, transport policy, inference governance and disclosure accounting. The cryptographic libraries encompass modern symmetric encryption, key derivation, hashing, signatures, post-quantum identity paths, zero-knowledge proof primitives, homomorphic encryption and differential privacy. Large cryptographic objects remain outside the compact semantic ABI and are referenced through fixed-size records.

DIDs and verifiable credentials provide portable identifiers and evidence of roles or qualifications. Delegated access and capability profiles are intended to ensure that a clinician, guardian, QApp or AI agent receives only the authority required for the purpose. Provenance and append-only records create the basis for accountability, while bilateral networking and publication classification distinguish private exchange from material intended for a commons. [E8][P1]

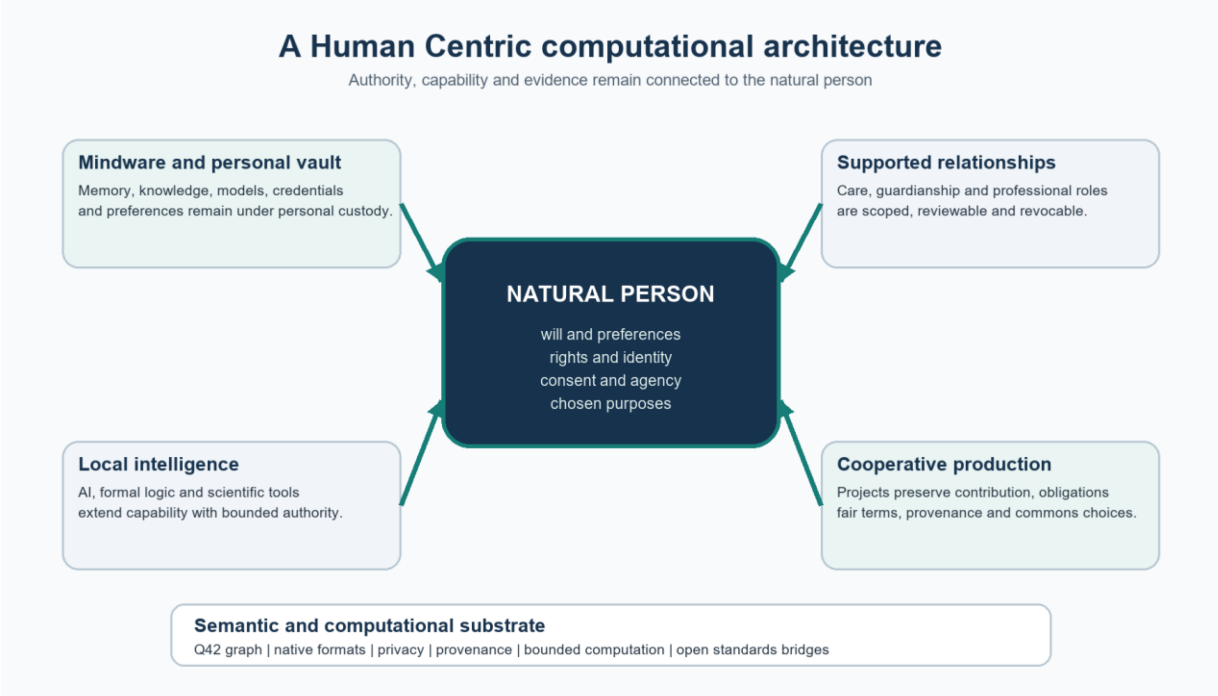

Figure 1. Authority starts with the natural person; the technical substrate exists to extend capability while returning evidence and accountability.

20. Interoperability, export and application surfaces

A Human Centric platform must meet people where they already work. Webizen Desktop and Studio provide panes for chat, ontology building, health and welfare, modelling, visualization and QApps. The CLI supports ingestion, query, profiles, model and daemon workflows. WebAssembly builds support browser-local logic and query. Solid/LDP compatibility supports institution-facing exchange. MCP-style tool interfaces expose bounded capabilities to agents. Declarative QApp manifests make applications portable and capability-scoped rather than inseparable from an app store.

The platform is therefore best understood as personal computing infrastructure: a semantic operating environment for information and action. Applications are views and instruments over the person's graph, authority and projects; they do not become the owner of those resources merely because they provide an interface.

Interoperability is evaluated at four levels. Semantic interoperability preserves identifiers, meaning and context. Syntactic interoperability provides standard serializations and query results. Operational interoperability lets Solid clients, browsers, command-line tools, QApps and agents perform bounded actions. Rights interoperability preserves consent, sensitivity, provenance, licence and purpose when information crosses a boundary. A successful export is not merely a readable file; it is a usable representation whose authority and obligations remain intelligible.

The Solid bridge illustrates the boundary design. HTTP, Turtle and JSON-LD are accepted in a network-facing component, constrained for size and translated into fixed hashes before records cross into the core VM. The core does not need to become a string-heavy web server in order to interoperate with Solid pods and LDP containers. Conversely, the person can disclose an authorized projection without exposing the packed internal volume or unrelated sanctuary material.

Application portability follows the same model. A QApp describes views, data bindings and requested capabilities; the host resolves those requests against the person's graph and authority. The same vault can therefore be presented through desktop, mobile, browser, CLI, accessible two-dimensional or immersive interfaces without redefining the person as an account belonging to any one application.

Part IV - Intended human journeys and measurable outcomes

21. Four illustrative journeys

Journey A - A person coordinating health and welfare

A person imports activity and sleep records, adds medications, attaches clinical documents, records housing and welfare concerns, and links these items to evidence and sensitivity classes. Local queries and clinical-support formulas help prepare questions for a health professional. A selected care summary is shared under a time-bounded usage agreement; inward notes and disputed material remain private. The useful outcome is not an automated diagnosis, but a more coherent account, better preparation, continuity across services and stronger participation in decisions.

Journey B - Supported decision-making with accountable guardianship

A person anticipates periods in which communication or executive function may be impaired. They define preferred supporters, separate medical from financial authority, specify triggers, record communication preferences and require two-of-three approval for specified high-risk actions. When support activates, every action cites its authority and evidence. The person can review the record, restore ordinary authority and dispute misuse. The outcome is continuity of agency through support, not replacement of the person.

Journey C - Building personal mindware for useful work

A learner or independent researcher ingests papers and notes into a semantic library, builds an ontology, queries relationships, runs statistics or simulation, asks a local model to help draft explanations, and uses symbolic tools to verify calculations. Sources, transformations and confidence remain attached. The work can be exported in open formats or packaged as a QApp. The outcome is expanded capability, retained privacy and attributable authorship.

Journey D - A cooperative project with fair contribution

Several people create a project DID, define roles and governance, plan work, record effort and capital, use shared ontologies and agents, and preserve contribution provenance. Obligations and payment terms are expressed before publication. Disputes enter review rather than being silently overwritten. Once an agreed cost or return threshold is met, the resource can move into a permissive commons under its licence. The outcome is collaborative production with a legible path from labour to recognition, compensation and shared benefit.

22. An outcome framework

The project can be evaluated by whether it increases practical human capability while reducing dependency and rights risk. Metrics should be interpreted with qualitative evidence and user control; the platform should not convert these measures into a hidden score of the person.

Outcome domain	Questions for evaluation	Illustrative indicators
Agency	Can the person understand, direct, refuse and reverse consequential actions?	Explicit purpose coverage; revocation success; correction time; user-understood authority paths
Health & welfare	Does the system improve continuity, preparation and access to support without overclaiming?	Record completeness; successful handoffs; unanswered-needs visibility; user-reported preparedness
Privacy & safety	Is sensitive use minimized, classified and auditable?	Local-processing share; blocked unauthorized disclosures; time-to-revoke; exposure surface
Supported agency	Do support arrangements preserve will and preferences?	Scope proportionality; review frequency; supported vs substituted decisions; successful restoration
Knowledge quality	Are claims attributable, contextual and uncertainty-aware?	Provenance coverage; contested-claim preservation; grounding quality; reproducibility
Useful participation	Can people create, analyse and coordinate rather than merely consume?	Projects completed; tools used; skills developed; reusable artefacts produced
Cooperation	Can groups coordinate without losing minority evidence or contributor identity?	Contribution attribution; dispute resolution; governance participation; cross-tool portability
Economic fairness	Are terms legible and contributions compensated on fair, reviewable bases?	Payment timeliness; obligation traceability; equal-value review; commons transition outcomes
Accessibility & plurality	Can people participate across language, disability, device and connectivity contexts?	Accessible modality coverage; language support; edge/offline completion; accommodation success

23. Design principles for continued development

- Keep the natural person as principal. Every agent, institution, model and app must expose the authority by which it acts.
- Protect selfhood by default. Biometrics, intimate memory, conscience and identity-linked mindware receive the strongest local and disclosure controls.
- Use the least powerful adequate authority. Scope, purpose, time and domain should be explicit and no broader than necessary.
- Preserve ambiguity when reality is ambiguous. Contradiction, uncertainty and minority evidence should remain representable.
- Separate observation, assertion, inference and decision. Provenance and epistemic status should travel with each transformation.
- Prefer local and deterministic tools for consequential work. Use generative AI as an assistant within a governed toolchain.
- Design for correction, revocation, portability and remedy from the beginning, not as export features added later.
- Treat accessibility, cultural and linguistic plurality as semantic requirements, not interface decoration.
- Make contribution and economic obligation legible while resisting perpetual extraction and enclosure.
- Measure benefit at the person's level: improved capability, continuity, participation and dignity - not platform engagement.

FORMAT ACCOUNTABILITY Treat fixed layouts, checksums, versions, migration paths and export mappings as public promises. Performance-oriented formats should remain inspectable and governed rather than becoming undocumented dependencies.

GRACEFUL DEGRADATION Every public or shared capability should provide a useful standards-based and accessible representation when specialized immersive, GPU or native-format support is unavailable.

SEPARATION OF MEANING AND DENSITY Keep rights, provenance, identity and policy in the semantic control plane; keep large model, media and geometry payloads in validated content-addressed containers joined by explicit references.

The format strategy expresses the project's wider balance: local sovereignty without isolation, high-performance native computation without opaque authority, and open exchange without forcing the core to abandon bounded execution. Q42, P64, Tensor10D, HCF/HMC, QApps and related manifests are intended to make advanced personal computation feasible; RDF, SPARQL, Solid and standard projections keep the resulting knowledge and work connected to a wider civic and technical world.

Conclusion

QualiaDB / Webizen is being designed as a Human Centric AI and personal computing platform whose technical ambition is matched by a normative one. The project seeks to give people a durable place for their data, identity, mindware, relationships and projects; a plural reasoning environment for difficult questions; advanced local computation for useful work; and a cooperative fabric in which authority, contribution and economic terms can be made explicit.

Its most distinctive feature is the attempt to integrate these concerns. Health support is connected to privacy and welfare. Guardianship is connected to consent, capacity and accountability. AI is connected to provenance and formal logic. Scientific modelling is connected to open semantic context. Cooperative production is connected to authorship, obligation and commons transition. The result is not simply a more capable database: it is an architecture for making computation answerable to the life, rights and purposes of the person using it.

> **THE PROJECT'S NORTH STAR** Advanced technical support should leave the person more capable, more informed, more connected on chosen terms, and better able to shape their own future - never less visible than the system that claims to assist them.

Sources and project evidence

Project sources were used to identify the architecture and intended design. External sources anchor the Maslow and human-rights framing. References describe the conceptual basis of this report; they are not an assurance of legal, clinical, security or production readiness.

Project sources

[P1] README.md; docs/manuals/ARCHITECTURE.md; AGENTS.md - platform thesis, semantic engine, local inference, logic, privacy, geometry, identity and human-facing surfaces.

[P2] docs/manuals/qualia_db_functionality_manual.md; crates/wellfare-core/src/ - WellFair health, welfare, finance, records, assessment, guardianship and project domains.

[P3] docs/manuals/webizen_permissive_commons.md; core-ontologies/permissive-commons.n3; core-ontologies/social-permissive-commons.n3 - publication classes, bilateral exchange, obligation costs and commons design.

[P4] crates/qualia-cooperative-core/src/; crates/wellfare-core/src/guardianship.rs; crates/wellfare-core/src/anatomy/birth.rs - supported agency, triggers, delegation, provenance and plural stewardship.

[P5] core-ontologies/values.n3; core-ontologies/selfhood.n3; core-ontologies/agency.n3; core-ontologies/jural.n3 - natural person, selfhood, personhood, roles, artificial agents and responsibility.

[P6] crates/qualia-core-db/src/DIRECTORY_INDEX.md; crates/qualia-core-db/src/modalities/; crates/qualia-core-db/src/specialized_libs/ - technical capability families.

[P7] docs/manuals/standards/q42-format-internal-draft.md; p64-weight-container-standard.md; q42-10d-tensor-standard.md; geometry-asset-ontology.md - native semantic, model and dense-asset formats, validation, versioning and execution boundaries.

[P8] docs/manuals/standards/qualia-vault-manifest.md; yaml-ld-q42-specification.md; hypermedia-content-format-hcf.md; docs/manuals/qapps_specification.md - portable vault, workspace, application and hypermedia formats.

[P9] docs/standards/qisp/0.1/profile.md; docs/manuals/adr/006-zero-allocation-solid-bridge.md - standards-compatible dense computation and the bounded Solid/LDP translation boundary.

External sources and standards

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[E2] United Nations. Universal Declaration of Human Rights (1948). Source

[E3] United Nations. International Covenant on Economic, Social and Cultural Rights. Source

[E4] United Nations. Convention on the Rights of Persons with Disabilities, especially Article 12; Committee General Comment No. 1 on equal recognition before the law. Source

[E5] UN Committee on Economic, Social and Cultural Rights. General Comment No. 23 on just and favourable conditions of work. Source

[E6] W3C. RDF 1.2 Concepts and Abstract Data Model (Candidate Recommendation Snapshot, 2026). Source

[E7] W3C. SPARQL 1.1 Query Language. Source

[E8] W3C. Decentralized Identifiers (DIDs) v1.0. Source

[E9] W3C. Shapes Constraint Language (SHACL). Source

[E10] Solid Community Group. Solid Protocol. Source